TIME SERIES FOUNDATION MODELS

# Sundial: A Family of Highly Capable Time Series Foundation Models

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Proceedings of the 42nd International Conference on Machine Learning (ICML 2025), PMLR 267



Figure 1 placeholder — We present native, flexible, scalable generative forecasting with Sundial

## Abstract — Key Ideas

- We present Sundial: native, flexible, scalable time series foundation models.
- We propose TimeFlow Loss: a flow-matching objective for continuous-valued forecasting without discrete tokenization.
- We pre-train on arbitrary-length inputs and generate multiple plausible futures.
- We make minimal but crucial Transformer adaptations and curate TimeBench with 1 trillion time points.
- We achieve state-of-the-art zero-shot point and probabilistic forecasting with millisecond latency.



## Introduction & Motivation

- We recognize forecasting is non-deterministic; decisions need distributions, not points.
- We observe many time series FMs are non-generative or rely on restrictive parametric densities.
- We find discrete tokenization causes OOV, coarse intervals, and long contexts.
- We aim for native continuous modeling, flexible distributions, and scalable training/inference.

# Related Work — Overview

### FORECASTING

- We position statistical vs deep models; we build foundation models for zero-shot generalization.
- We note probabilistic forecasting with parametric losses risks mode collapse.

### TIME SERIES FMS

- We summarize TimesFM, Timer, Time-MoE: continuous embeddings + MSE/quantile.
- We compare Moirai (mixtures) and Chronos (discrete quantization with cross-entropy).

### GENERATIVE MODELING

- We contrast diffusion vs flow-matching in continuous modalities.
- We require and develop a conditional, autoregressive formulation for forecasting.



# Preliminaries — Flow-Matching

- We learn a time-dependent velocity field to transport a source Gaussian to the target data distribution.
- We optimize the conditional flow-matching objective; this is efficient vs diffusion.
- We sample via a push-forward ODE using the learned velocity.

Equations/ODE path placeholder — conditional flow-matching pipeline



# Probabilistic Forecasting Setup

- We predict p(x[t+1:t+f] | h_t) with h_t from a Transformer.
- We use a small conditional generative head to draw samples, then compute median/quantiles.
- We model uncertainty beyond fixed prediction intervals.



# Sundial Architecture

- We apply stationarization and patch embedding for arbitrary-length series; we keep original continuous values with padding/masks.
- We use a decoder-only Transformer with Pre-LN, RoPE, FlashAttention, and KV cache for stability and speed.
- We optimize TimeFlow Loss for per-token predictive distributions and fast sampling.
- We use multi-patch prediction to reduce autoregressive steps and reuse lookback representations for rapid multi-sample inference.

Figure 2 placeholder — Input → Patch tokens → Transformer → TimeFlow head → Samples

## Evaluation Datasets

- We evaluate on TSLib (ETTm1, ETTm2, ETTh1, ETTh2, ECL, Weather).
- We evaluate on GIFT-Eval (23 datasets).
- We evaluate on FEV (27 datasets).

Figure placeholder — Benchmarks used for evaluation

We use widely recognized benchmarks



## Results — Point and Probabilistic

### POINT FORECASTING (TSLIB)

- Consistent SOTA zero-shot on ETTm1/2, ETTh1/2, ECL, Weather.
- ~7.57% lower MSE and ~4.71% lower MAE vs Time-MoE with fewer parameters.
- Patch-level forecasting reduces autoregression steps vs point-wise tokenization.

### PROBABILISTIC (GIFT-EVAL & FEV)

- GIFT-Eval: 1st in MASE, 2nd in CRPS among 97 configurations.
- FEV: zero-shot exceeds ~70% supervised/statistical models; ~35× faster inference vs Chronos.
- Median/quantiles computed from raw samples without prior densities.

Strong accuracy and calibrated uncertainty, zero-shot



## Limitations and Calibration

### LIMITATIONS

- Zero-shot can degrade on unseen domains; broad data coverage helps.
- Pre-training and multi-sample inference are compute-intensive.
- Miscalibration with too few samples/steps; priors may over-smooth.
- Univariate pre-training with per-variable normalization may miss cross-variable dependencies.

### TEST-TIME CALIBRATION

- Increase number of samples to stabilize median/quantiles.
- Use finer K push-forward steps for precision.
- 20 samples × 50 steps ≈ ~1s on CPU; no retraining needed.

Practical trade-offs: data, compute, and post-hoc control



## Objectives vs Adaptation — Comparison

### TRAINING OBJECTIVES

- TimeFlow outperforms diffusion and MSE on average MSE (TSLib).
- Diffusion underperforms notably; MSE yields single deterministic, often over-smooth outputs.
- CRPS favors TimeFlow for coherent, diverse predictive distributions.

### MODEL ADAPTATION

- Fine-tune Sundial (Base) once on aggregated FEV tasks.
- Improves over zero-shot; training from scratch underperforms → transfer from pre-training.

Efficient objective and effective transfer

# Scalability

- Larger Sundial models yield better downstream results.
- ~15.38% lower converged training loss on TimeBench (Large vs Small).

Figure placeholder — Training curves (Small vs Large)



# Contributions

- We introduce TimeFlow Loss: generative, continuous, tokenization-free probabilistic forecasting.
- We build the Sundial family: scalable Transformers with deployment-ready adaptations.
- We curate TimeBench and achieve SOTA zero-shot across point and probabilistic benchmarks.

SOTA Generative Scalable